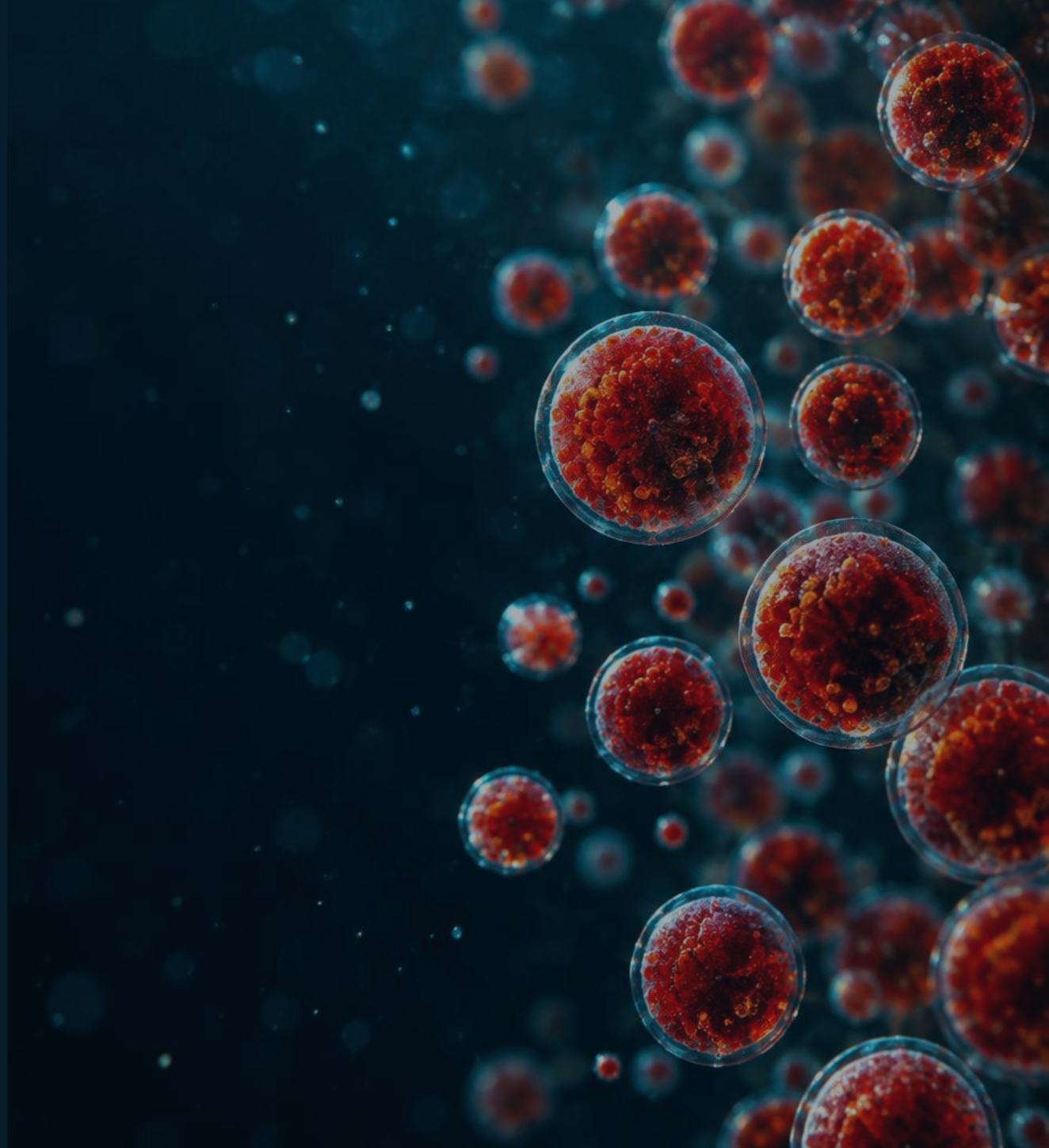
MARINE ALGAL BIOTECHNOLOGY

Astaxanthin Synthesis and Production

From biosynthetic pathways to scalable bioprocessing

Prof (Assoc) Wa Iba
ASMI Webinar, 30/10/20



Learning objectives & production landscape

TRACE

Connect carbon metabolism to carotenoid synthesis.

DESIGN

Relate stress, cultivation and downstream choices.

EVALUATE

Separate evidence, economics and scale-up claims.

Scope note

Haematococcus pluvialis is a freshwater chlorophyte and the commercial benchmark—not a marine alga. We use it as a process reference while distinguishing marine-derived sources, accumulators and true biosynthetic producers.

What is astaxanthin?

C_{40}

ketocarotenoid · xanthophyll

Conjugated polyene chain

Delocalized electrons drive visible absorption and radical-quenching behavior.

Polar terminal rings

Hydroxyl and keto groups shape membrane orientation and reactivity.



In biomass, astaxanthin is often mono- or di-esterified with fatty acids; analytical methods must report free versus esterified pools.

Biosynthesis: carbon → carotenoid

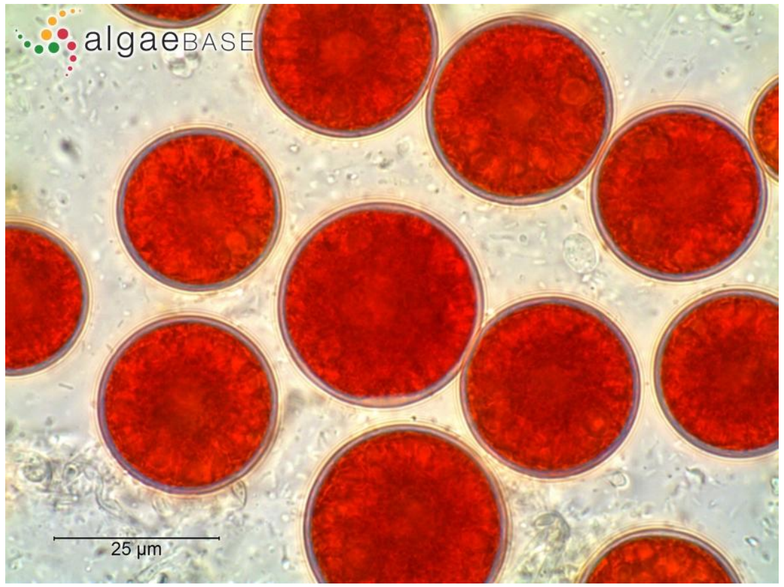


Branching and flux control matter

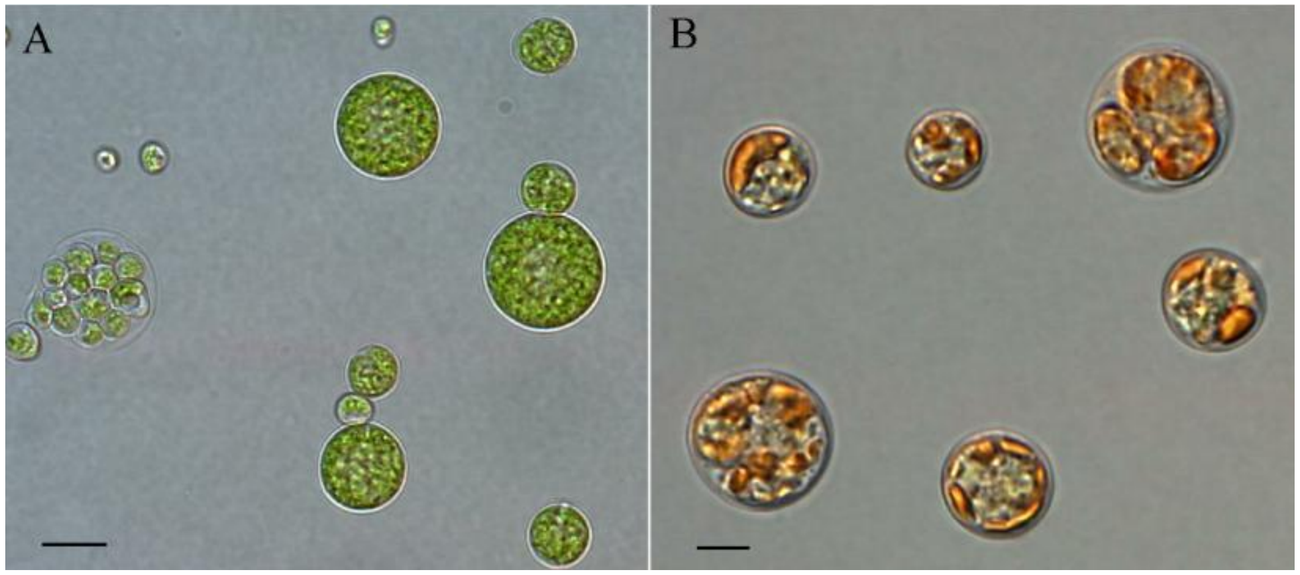
β-carotene can feed competing carotenoids; enzyme order and organism-specific isoforms vary. This is a teaching map, not a universal stoichiometric scheme.

Natural sources ≠ biosynthetic producers

ORGANISM	HABITAT	ROLE	PRODUCTION RELEVANCE
Haematococcus pluvialis	Freshwater	Synthesizes + accumulates	Commercial benchmark
Chlorella zofingiensis	Freshwater / tolerant	Synthesizes	Engineering candidate
Phaffia rhodozyma	Terrestrial yeast	Synthesizes	Fermentation
Paracoccus carotinifaciens	Bacterium	Synthesizes	Feed pigment
Marine animals / biomass	Marine	Often accumulates via diet	Source, not proof of synthesis



Haematococcus pluvialis (iSource : Algaebase.org)



Light micrographs of *C. zofingiensis* cells under favorable (A) and stress (B) conditions. Bars, 5 µ m (Source: Liu et al., 2014)

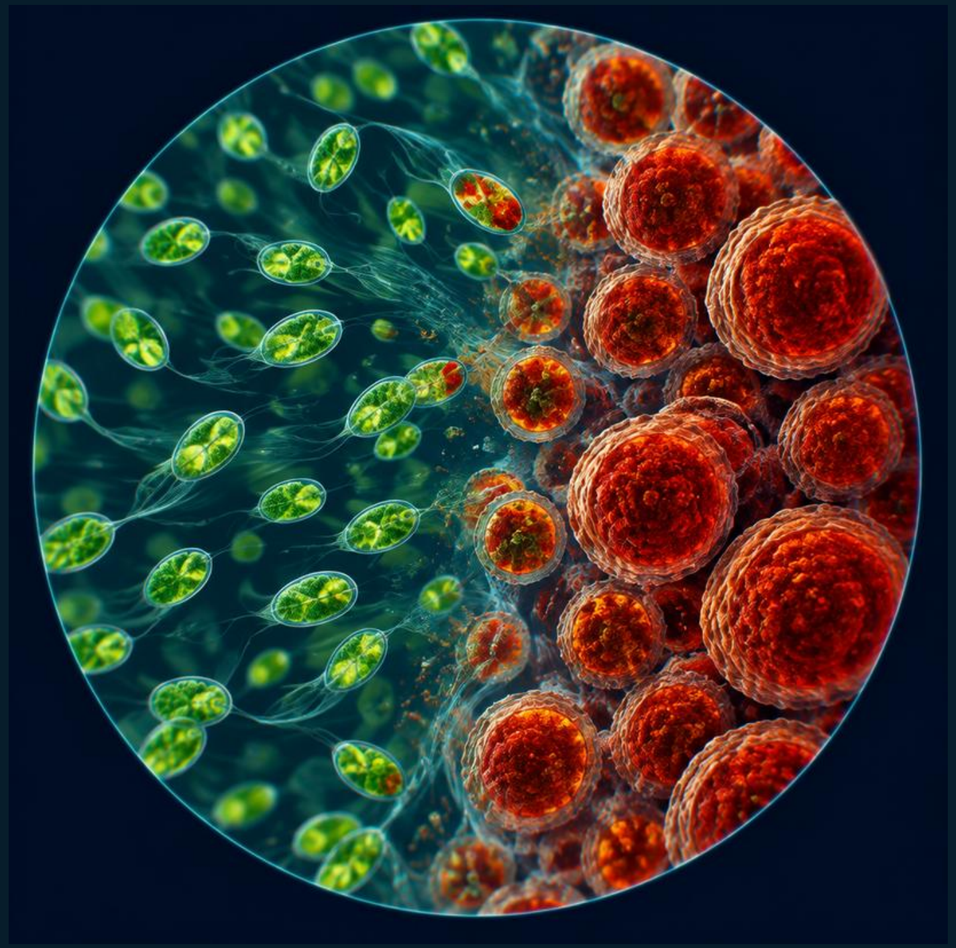
The green-to-red two-stage process

01 GROW

Favorable light, nutrients and carbon support motile green cells and biomass build-up.

02 STRESS

High light plus nutrient and/or osmotic stress drives non-motile cyst formation and pigment accumulation.



Bioprocess controls: stress is combinatorial

LIGHT

Intensity · spectrum · photoperiod

CLIMATE

Temperature · pH · salinity

NUTRIENTS

N/P limitation · trace metals

CARBON

CO_2 delivery · organic carbon

BIOLOGY

Strain · stage · inoculum history

TIME

Dose · sequence · recovery window

Optimize combinations—not isolated maxima. Productivity = biomass × cellular content ÷ time.

Cultivation platforms and scale-up

CLOSED PBR

Tubular / flat-panel

+ contamination control
+ defined gas transfer
− light and O_2 gradients
− cooling, fouling, high CAPEX



OPEN RACEWAY

Ponds / channels

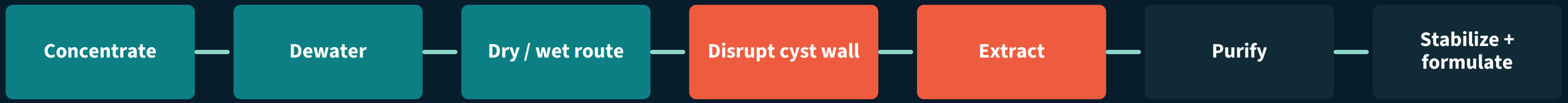
+ lower infrastructure cost
+ simpler large footprint
− contamination / evaporation
− weaker environmental control



Scale-up is a transport problem: photons, gases, heat, water and cells must all move efficiently.

Downstream processing: protect value at every step

UNIT OPERATIONS



Centrifugation · filtration · bead milling · high-pressure homogenization · solvent or supercritical CO_2 extraction

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Design rule

Cell-wall severity, moisture, solvent polarity and oxygen exposure determine recovery—not extraction yield alone.

Product quality, stability & formulation

PRIMARY RISKS

O_2

LIGHT

HEAT

PROTECTIVE DESIGN

Low-oxygen processing
Opaque / barrier packaging
Encapsulation or lipid carriers
Antioxidant-compatible matrices

QUALITY CONTROL

HPLC profile
Isomer ratio
Free vs esterified
Oxidation markers
Residual solvents

Formulation is part of the bioprocess—not an afterthought.

Applications and evidence boundaries

ESTABLISHED

Food & feed pigmentation
Aquaculture coloration
Ingredient antioxidant claims within regulation

EMERGING

Nutraceutical outcomes
Cosmetic performance
Biomedical mechanisms and clinical translation

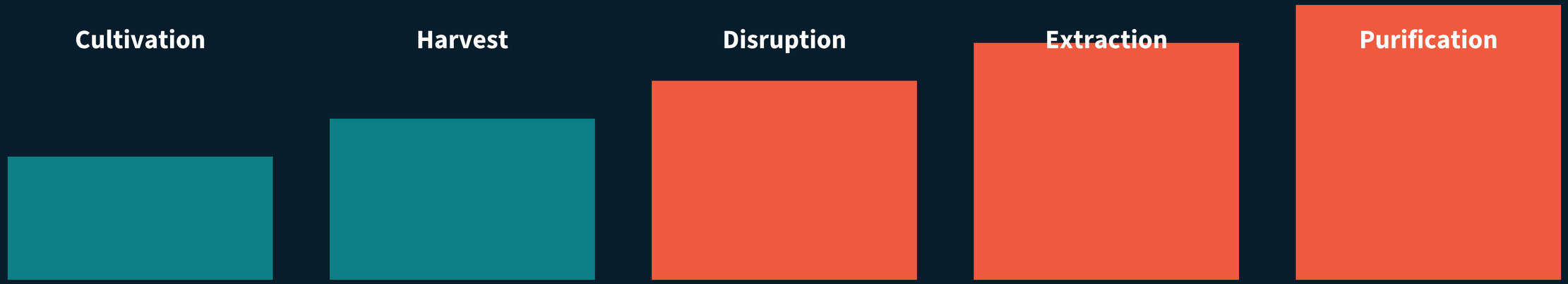
Evidence ladder — **Mechanism → preclinical → controlled human evidence → approved claim**

Do not infer clinical efficacy from antioxidant chemistry alone.

Techno-economics & sustainability: where value leaks

HISTORICAL CASE STUDY

Panis & Rosales Carreon (2016) estimates are scenario-specific historical values—not current market prices.



Design levers

Wastewater / low-cost nutrients · water and energy recovery · coproduct valorization · solvent recirculation · life-cycle assessment

Emerging directions: a research roadmap

1	ENGINEER	CRISPR and pathway balancing
2	INTENSIFY	Smarter light / nutrient delivery
3	GREEN	Safer solvents and milking extraction
4	INTEGRATE	Wastewater + coproduct biorefineries

CAVEATS

Off-target effects
Regulatory burden
Nanomaterial fate
Continuous recovery losses
Variable wastewater

Three takeaways

01 Name the biology correctly

Freshwater benchmark ≠ marine source.

02 Optimize the whole chain

Flux, stress, transport and recovery interact.

03 Bound claims with evidence

Technical promise is not clinical or economic proof.

SELECTED REFERENCES

Shah et al. (2016) Trends Biotechnol. 34: 329–341. https://doi.org/10.1016/j.tibtech.2015.12.001

Panis & Rosales Carreon (2016) Algal Research 18: 175–190. https://doi.org/10.1016/j.algal.2016.06.013

Ambati et al. (2014) Mar Drugs 12: 128–152. https://doi.org/10.3390/md12010128

Torres-Tiji et al. (2020) Biotechnol Adv 41: 107536. https://doi.org/10.1016/j.biotechadv.2020.107536